Title pending

Antoni Nikolov Catalin Popoiu Guilherme Oliveira

Abstract. Nothing here yet

1 Methods

The objective of this research is to interrogate a variety of methods within the literature surrounding time-series regression. More specifically, energy price or energy related time-series prediction. This particular research field adopts a multi-feature, and often multimodal approach to the problem. This section outlines the methods used for the research, in hopes of finding a model architecture that outperforms the current method that OpenRemote employes.

1.1 Research Design

The research is designed to almost perform as a census on the available methodologies. A healthy variety of methods for preprocessing and architectures were choosen and divided into experiments that compile similar architectures, with sub experiemnts which introduce small variations. These variations intend to reveal the effect of a particular method i.e. if the difference between experiment 1.a and 1.b is one specific architecture, then the difference in performance might have to do with the introduction of that architecture.

The scope of the research is strictly for time-series energy price prediction, and its goal of being more accurate than the benchmark.

We will, for the purposes of this proposal, use a naming convention of Exp.$i.v$, where i is the number of the experiment configuration, and v is a letter, whose order is reset for each i.

1.1.1 Exp.1

From our literature, one of the most recurrent topics is the power of ensemble models, particularly ones using RNNs. To start out simple, we will introduce a CNN-LSTM variant architecture as a base and design multiple experiments that build upon it.

Since we are using CNNs, which serve the purpose of capturing long-term sequential patters, it would be ideal to test an array of preprocessing methods, some of which enhance 3-dimentional features. For Exp.1, all variations will cycle through:

1. Normalisation
2. Wavelet Transform[9, 2]
3. Transformer Encoding (PatchTST[7])

Exp.1.a Variation **a** will use a simple CNN-BiLSTM, since the use of BiLSTM consistently outperforms normal LSTMs in time-series forecasts.

Exp.1.b Variation **b** will use a CNN-xLSTM.

Exp.1.c Variation **c** will use a CNN-BiLSTM-Transformer. The discrepancy between results for each preprocessing method should be greatly scrutinised for this variation, as the use of a Transformer encoder and output model essentially places the CNN and BiLSTM as feature extractors.

Exp.1.d Variation **d** will use a CNN-Transformer. By removing the transformer, we aim to prove the null hypothesis that RNNs, particularly LSTMs, are greatly outperformed by Transformers for short-term dependencies, and that the features that they extract act as noise for a transformer.

This set of variations offer a comprehensive use of the methods that the literature has provided, but they also take into account standard practises i.e. Normalisation and the use of CNNs for time-series data, also found in the literature[1, 12, 6].

1.1.2 Exp.2

Current trends in AI strongly favour Transformer architectures, or Attention layers. Exp.2 will focus on exploring this particular aspect in more detail. Exp.1 also features Transformers, but in

	Exp.1			
	a	b	c	d
Architecture	CNN-BiLSTM	CNN-xLSTM	CNN-BiLSTM-Transformer	CNN-Transformer
Preprocessing methods	Norm., WT, PatchTST	Norm., WT, PatchTST	Norm., WT, PatchTST	Norm., WT, PatchTST
Additional notes				To prove null-hypothesis i.e. RNNs punish accuracy

Table 1: Exp.1: Model Architectures and Preprocessing

	Exp.2				
	a	b	c	d	e
Architecture	Decoder-Only Transformer	Decoder-Only + FFT	Encoder-Decoder Transformer	Kernel-based Transformer	iTransformer
Preprocessing methods	Standard	Wavelet Transform	Standard	Kernel-based	Standard
Additional notes		Same model as **a**			

Table 2: Exp.2: Model Architectures and Preprocessing

this section, we want to amplify their strenghts by means of expanding our number of features. Non-linearity is accutely endemic to all of our features[6]. One hypothesis is that the Transformer architecture will be able to capture linear features where there is usually non-linearity.

Exp.2.a Variation **a** will use a classic multi-head Decoder-Only transformer[8]. Decoder-Only transformers perform well in renewable scenario analysis, and use less memory[3]. This is relevant for our study because the nature of the data and analysis is similar by virtue of the non-linear and complex time-series tasks both try to address.

Exp.2.b Variation **b** will use the same model as Exp.2.a, but utilise Wavelet Transformers for feature extraction[13].

Exp.2.c Variation **c** will use a multi-head Encoder-Decoder Transformer. Encoder-Decoder variants perform great with forecasting tasks[3].

Exp.2.d Variation **d** will use a Kernel-based Transformer. This architecture can better capture linearity when it exists[10] which has been pointed out as a reason for transformers failing to generalise on Time-Series tasks[4] , all the while using less memory.

Exp.2.e Variation **e** will use an inverted Transformer, henceforth known as iTransformer; a method introduced by a landmark paper back in 2023[5]. The paper specifically demonstrates the effectiveness of the proposed methodology for Time-Series Forecasting, and has been explicitely used for EPF[11].

1.2 Data

The dataset comprehends a meryad of features from different sources, such as metereological data, overall energy consumption and production, and gas prices. The target variable, being the energy price, is also included. The $\vec{\Delta}$ between samples is 1 hour.

The data is also preprocessed using the methods above. Timestamp data is feature engineered to decompose it into many features, such as hour, day of week, month, quarter and weekend, all with cyclical scaling.

1.3 Procedure

All experiment models are based on the Torch library. For the preprocessing, there is a mixture of self-written methods, and some from scikit-learn and pywt, for Wavelet transform. They are arranged into the experiment configurations described above, and run sequentially using a custom script, that outputs the results in bulk.

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